

Social Support is Contagious: Exploring the Effect of Social Support in Online Mental Health Communities

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Abstract

It is known that social support is beneficial for mental health. With the development of social media, social support could also influence the development of online mental health communities. However, few studies examined these effects from the perspective of online mental health communities. This study focused on 22 mental health related subreddits and conducted causal analysis by matching and comparing users (1) who received social support with who did not receive social support; (2) who received positive social support with who received negative social support. The results shows that social support is “contagious”: users who received social support on their first post would be more likely and quickly to post again and provide support for others; users who received positive support would also provide more positive social support for others in the future. Our findings indicate the potential chain reaction of social support. This study also provides insights into how online mental health communities can better facilitate people to spread social support and deal with mental problems.

CCS Concepts

• Human-centered computing → Collaborative and social computing.

Keywords

Social Media, Online Community, Social support, Mental Health, Reddit

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1 Introduction

It is known that social support can positively affect psychological well-being [17] and reduce mental health problems, e.g., depression

[22, 32, 38], stress [6], and mental disorder [10]. Research has explored the mechanism in which social support works [18, 35] and demonstrated the impact of social support on different populations, such as college students [15], elderly people [21], and low-income populations [3]. With the rapid development of social media, online mental health communities (OMHCs) are widely used for mental health purposes [4, 20, 33] and are regarded as an important channel to understand people’s mental state [7, 25]. Considering the usual structure of OMHCs, research views the posts on mental health communities as seeking support and the comments as providing support [11, 12]. Experiments have been widely used to study the effect of online social support [22, 38]. Research also considered the ethical aspects of relevant experiments and explored methods of causal analysis based on social media data. The results of these studies all showed the positive effect of social support on individuals’ mental health state [11].

In online communities, users are not only the recipients but also the providers of social support. Their willingness to provide social support is essential for the development and success of online communities [5, 37]. Therefore, in addition to the impact of social support on mental health, studies also examined its impact on user engagement and community development. Qiu et al. [28] found that 75% of users who received at least one reply from others would subsequently express positive sentiment. Cunha et al. [9] focused on a weight-loss community and found that users receiving more positive comments on their initial post were more likely to return to the community in the future. Lampe and Johnston [19] examined a news and discussion site and found if a users received a negative feedback at the first post, it might take them longer to post again. Interviews to online users also found that positive social support boosted subsequent feelings of pride, self-confidence, and status, and motivated content contribution [40].

Although a rich body of work has shown the positive effects of social support on people’s mental health state and user contributions, few studies examined these effects from the perspective of online mental health communities, which has become a crucial resources for online mental health support [31]. Therefore, our work seeks to address the research gap by focusing on OMHCs in Reddit and examining the impact of social support on individuals’ subsequent behaviors. Specifically, this study aims to answer the following research questions.

RQ1: Does received social support at the first post affect users’ probability of returning to online mental health communities?

RQ2: Does received social support at the first post affect users’ return speed to online mental health communities?

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RQ3: Does received social support at the first post affect the sentiment of users' subsequent self-disclosure and support that they offer to others in online mental health communities?

We conducted causal analysis by matching similar users who did with who did not receive social support at their first post and comparing their probability and speed to provide social support to other people and the sentiment of their following published content. We further applied the same approach to analyze the different impact of positive and negative support on users' subsequent behaviors. We found evidence that receiving social support from others at the first post contributed to higher probability to post again and offer social support to others in online mental health communities; it also reduced users' return speed. Additionally, we found that users who received more positive support will subsequently provide more positive support for others. This study suggests the "contagious" feature of social support. Our findings provide insights into the design of OMHCs to make them play a positive role in dealing with mental health problems and help them achieve long-term and healthy development.

2 Methods

2.1 Data Collection

Reddit is a widely used online forum where users can create a post or comment on existing posts. The posts are organized by their topics into a variety of communities called "subreddits" [31]. Prior research obtained a list of mental health related subreddits using snowball approach [26]. In this study, we focused on these 22 mental health related subreddits (also referred to as OMHCs)¹.

In order to examine the effect of social support at the first post, we selected users who satisfied two requirements: (1) Has not posted in any OMHCs from September 2016 to August 2017; (2) Has posted in at least once OMHC from September 2017 to August 2018. The first requirement was set to make sure the user has not been involved in OMHCs for a relative long period so his or her first post in (2) could be viewed as the first post in this study. We used Google's BigQuery to obtain all users satisfying above requirements and collected their first post and attached comments. We also collected all their subsequent posts and comments within a year after the first post. Since social support is an interpersonal behavior, this study only considered the comments that were made by users except the original poster. This provided us with 105,739 users, 105,739 first posts and 487,877 attached comments, 99,683 subsequent posts, and 435,041 subsequent comments.

2.2 Propensity Score Matching

The simplest way to analyze the treatment effect is to subtract the outcome variable when someone does not receive treatment from that when someone receives treatment. However, we can not observe both results of being and not being treated in reality. Therefore, we matched similar pairs in which one received treatment while the other did not, then compute their difference to analyze the

effect of treatment. Propensity score matching (PSM) [29] method matches users according to their probability to receive treatment and it attempts to reduce the bias due to confounding variables [36]. In this study, we adopted PSM to matched users and examined the treatment effect.

2.2.1 Definition of Treatment and Control As comments are viewed as an important form of social support in online communities [11, 12], we defined one treatment factor in this study as "receiving at least one comments at the first post". That means the control group receive no comments on their first post. In this way, we obtained 64,514 treatment group candidates and 41,225 control group candidates.

In order to further examine the impact of comment sentiment, we matched users who received different sentiment of social support at the first post. Specifically, we defined the second treatment factor in this study as "positive average sentiment score of comments". We used VADER (Valence Aware Dictionary for Sentiment Reasoning) [16] score to measure the sentiment since it applies to social media data. We classified posts with a positive average score of comments into the treatment group, and classified posts with a negative average score of comments into the control group. This finally provided us with 43,056 treatment group candidates and 19,276 control group candidates.

2.2.2 Covariate Selection Since users' personal information such as demographics and images are not available in user profiles or other webpages in Reddit, the selection of covariates in our research is based on the hypothesis that posts with similar content will have a similar probability of receiving a comment [8]. Therefore, we chose the following variables about the first post:

- Word Count: Number of words in first post's content (title + subtext).
- 20 LDA topic score: We generated a topical representation of first post's content by Latent Dirichlet Allocation (LDA) [1]. The parameters are set according to prior research [8].
- LIWC features: Linguistic Inquiry and Word Count (LIWC) [27] is a well-validated psycholinguistic lexicon which has been widely used in prior research [23, 31]. In this study, we chose a set of 27 categories in LIWC that are suitable for mental health related content [34], e.g., anxiety, anger, and sadness.
- Empath features: We focused on 20 Empath [13] categories related to mental health to supplement LIWC features, e.g., nervousness, disappointment, and fear.
- VADER features: We used positive, negative, neutral and compound scores generated by VADER.
- Dummy variables for OMHCs: Considering the potential difference of 22 OMHCs, e.g., activity level, community rules, we created 21 dummy variables for these OMHCs.

We then adopted a logistic regression with LASSO (Least Absolute Shrinkage and Selection Operator) to further select variables for PSM from all above features. LASSO is a regression analysis method that performs both variable selection and regularizations. We computed the mean AUC (Area Under the ROC Curve) over 10-fold cross validation setting. For the first treatment factor, we finally obtained a model including 20 variables, in which AUC is

¹The list of mental health related subreddit: r/depression, r/mentalhealth, r/Anger, r/ptsd, r/traumatoolbox, r/psychoticreddit, r/SuicideWatch, r/MMFB, r/getting_over_it, r/survivorsofabuse, r/alcoholism, r/BPD, r/rapecounseling, r/bipolarreddit, r/addiction, r/DPDR, r/hardshipmates, r/feelgood, r/StopSelfHarm, r/panicparty, r/socialanxiety, r/EatingDisorders.

0.61. For the second treatment factor, we finally obtained a model including 13 variables, in which AUC is 0.60.

2.2.3 Matching and Validation We conducted propensity score matching to remove the relationship between the covariates and supposed causal variable. Based on selected variables, we conducted one-to-one nearest neighbor matching with a caliper of 0.01. If there are more than one matched users to a certain user, we selected the pair with smallest Mahalanobis distance. For the first treatment factor (received comment or not), we finally obtained 39,420 pairs of users. And for the second treatment factor (received positive or negative comments), we obtained 19,086 pairs of users.

We checked the absolute standardized mean difference (ASMD) of covariates before and after matching, the results showed that the ASMD of all covariates after matching were less than 0.1, which meant the treatment group and control group were well balanced [30].

3 Results

In this section, we compare the outcome variables of two matched groups to analyze the effect of social support. Prior research has shown the difference between posts and comments in OMHCs, that posts are published to seek support while comments are used to provided support [11, 12]. So we examine the user behaviors about posts and comments separately.

3.1 Return Probability(RQ1)

3.1.1 Received Comment or Not For the “comment group” that received at least one comment on the first post and “no comment group” that did not receive any comment on the first post, we first compared their probabilities of returning to the OMHCs to post or comment. Specifically, we considered the probabilities of the following four activities: (1) return to post in the same OMHC as the first post, (2) return to post in any OMHC, (3) return to comment in the same OMHC as the first post, (4) return to comment in any OMHC. As Table 1 shows, all these four return probabilities would increase significantly if users’ first post received at least one comment.

Variable	Comment Group	No Comment Group	Chi square
Post in the same OMHC	25.4%	20.3%	296.125***
Post in any OMHC	30.8%	25.6%	261.011***
Comment in the same OMHC	39.5%	25.7%	1699.358***
Comment in any OMHC	43.0%	29.0%	1669.398***

Table 1: Comparing the return probability of “comment group” and “no comment group”. * = $p < .05$, ** = $p < .01$, *** = $p < .001$.

The probability of returning to post in the same OMHC and in any OMHC in the “no comment group” was 5.1% and 5.2% higher than that in the “comment group” respectively. Since we have controlled the differences between the two groups in terms of sentiment, topics, and expression way of their first posts, the results suggest that the higher post return probability of the treatment group was caused by received comments of their first post. Prior

research has shown that users might post in OMHCs to seek social support [11, 12]. We randomly selected 1000 posts from our dataset and manually coded them. The results show that 94.7% of 927 unremoved posts include users sharing their personal experience and seek support. Therefore, our results in this section suggest that social support increases the probability of subsequent self-disclosure and support seeking behaviors in the OMHCs.

The difference in the probability of returning to comment between the two groups are even greater. Users in “comment group” were 13.8% more likely to comment to others in the same OMHC, and 14.0% more likely to comment to others in any OMHC after the first post. The results show that social support is “contagious”, which means receiving social support (comment) on the first post significantly increases probability of users providing others with social support in the future.

3.1.2 Positive Comment or Negative Comment For users received at least one comment, we further explored the effect of comment sentiment. As Table 2 shows, users received positive comments and negative comments had no significant difference in probability to return and post in the same OMHC as the first post, but those receive negative comments have even 1.0% higher probability to return and post in any OMHC. As for the commenting behavior, the return probability of the “positive comment group” was significantly 1.9% higher than that of the “negative comment group” both in the same OMHC as the first post or in any OMHC. These results suggest that positive social support are effectively in increasing the probability to provide others with social support, but may negatively influence the probability to return to seek help.

Variable	Positive Group	Negative Group	Chi square
Post in the same OMHC	25.1%	25.9%	3.487
Post in any OMHC	30.7%	31.7%	4.410*
Comment in the same OMHC	40.3%	38.4%	13.912***
Comment any OMHC	44.0%	42.1%	13.465***

Table 2: Comparing the return probability of “positive comment group” and “negative comment group”

3.2 Return Speed (RQ2)

3.2.1 Received Comment or Not For users who did return to OMHCs to comment or post, we further explored the impact of social support on the return speed, which is measured by the time between the first post and the following utterance (post or comment). As Table 3 shows, for users who posted again in any OMHC, there is no significant difference in return speed between “comment group” and “no comment group” ($p = 0.1258$). However, “comment group” returned to the same OMHC to post 3.9 days more quickly. This suggests that the impact of social support on return speed is limited to the OMHC of the first post.

When it comes to return speed of comment, however, the results were quite different. For users who returned to comment in the same OMHC as the first post, the mean return time of the “comment group” was 10.2 days less than that of the “no comment group”, and the difference was significant. And for those who returned to comment in any OMHC, the mean return time of the “comment

group” was 11.2 days less than that of the “no comment group”, and the difference was also significant. These results indicate the effect of support in another dimension. That is, receiving social support not only makes users more probable to subsequently offer others social support, it also increases the speed of that behavior, allowing them to return to the community more quickly as supporters.

Variable	Comment Group Mean(Num.)	No Comment Group Mean(Num.)	t statistic
Post in the same OMHC	53.7 (7219)	57.6 (5010)	-2.682**
Post in any OMHC	56.5 (12151)	58.3 (10109)	-1.531
Comment in the same OMHC	30.8 (6914)	41.0 (4726)	-7.944***
Comment in any OMHC	33.1 (8423)	44.3 (5939)	-9.356***

Table 3: Comparing the return speed (days) of “comment group” and “no comment group”

3.2.2 Positive Comment or Negative Comment As for the impact of comment sentiment, the results show that it has no significant effect on return speed (Table 4). That means although comment sentiment affect individuals’ probability to publish another post or comment, it does not influence how quickly they will do that.

Variable	Positive Group Mean(Num.)	Negative Group Mean(Num.)	t statistic
Post in the same OMHC	53.8 (3485)	53.1 (3569)	0.368
Post in any OMHC	55.9 (5850)	55.2 (6041)	0.483
Comment in the same OMHC	31.6 (3324)	29.8 (3394)	1.159
Comment in any OMHC	33.8 (4106)	32.2 (4180)	1.132

Table 4: Comparing the return speed (days) of “positive comment group” and “negative comment group”

3.3 Comment Sentiment (RQ3)

To better understand the impact of social support on individuals’ mental health state, we examined the influence of social support on sentiment expression by calculating the average VADER sentiment value (from -1: most negative to 1: most positive) of subsequent k ($k=1,2,3,4,5$) posts or comments after the first post.

3.3.1 Received Comment or Not First we compared the sentiment expression between those received social support and those did not receive social support. The result (Table 5) suggests that “comment group” and “no comment group” have no significant difference in sentiment of their subsequent posts or comments. That indicates even though whether receive social support or not influences users’ probability to return and the return speed, it does not contribute to users’ sentiment expression in the future.

3.3.2 Positive Comment or Negative Comment Then we explored the impact of comment sentiment on posters’ further sentiment expression. As Table 6 shows, there is significant difference between “positive comment group” and “negative comment group” on the sentiment expression of both subsequent posts and subsequent comments. “Positive comment group” will publish more positive posts and comments in the future, while negative comment group will publish more negative content. These results show that similar to the probability of providing social support, the sentiment of social support can also propagate.

3.4 Stability Analysis of Time Selection

In previous sections, we analyzed the effect of social support on individuals’ subsequent behaviors. However, all the analysis took one year as the research unit. That is, to focus on users who did not post for one year, and to see if they return to the OMHCs in the year after their first post. In order to investigate the stability of the results, we also tried other time periods. Specifically, we set the time period as 6, 3 and 1 month respectively and carried out the same analysis as previous sections. The results show that our findings are not sensitive to the choice of time period.

4 Discussion

4.1 Impact of Receiving Social Support

Our results indicate that users who received at least one comment from others would be more likely to post and comment in OMHCs. That is aligned with existing studies about the relationship between feedback and content contribution [2, 39], but provides new insights to the area of social support. Since we focused on OMHCs, our results showed the significant effect of social support on users with mental health concerns. Additionally, as prior research showed that posters in OMHCs act as support seekers and commenters in OMHCs act as support providers [11, 12], our results suggest that users who received social support on the first post in OMHCs would be more likely and more quickly to continue to disclose themselves and seek help, and give others support in the future. However, as for the impact of sentiment of subsequent posts and comments, we did not find significant difference between the “comment group” and the “no comment group”. In summary, this study suggests that whether or not an individual received social support had a positive effect on his future social support provision and self-disclosure behaviors, but it did not have significant effect on users’ subsequent sentiment expression in those behaviors.

4.2 Impact of Receiving Positive Social Support

To further explore the effect of the sentiment of social support on users’ subsequent behaviors, we matched users who received social support with an average positive sentiment and those with an average negative sentiment. For the return probability, our results are partly aligned with prior research that showed positive feedback encourage behaviors [24]. This study suggests that the “positive comment group” would return more quickly to provide social support for others, while the “negative comment group” would return to OMHCs more quickly to publish a post. That means compared to negative social support, positive social support could increase the probability to offer other people social support, and cause negative effects on the probability of seeking social support again. In addition, we found that the impact of the sentiment of social support is similar to “emotional contagion” phenomenon, which means that one person’s emotions and related behaviors may trigger similar emotions and behaviors of other people [14]. Our findings show that users who received positive social support would publish significantly more positive posts in the following posts, and also provide much more positive social support (comments) to others. All these results provide insights into the design of online mental health communities to develop successfully and better support their users.

Variable \ k	1	2	3	4	5
Post in the same OMHC	-0.001 (-0.227,-0.226)	-0.003 (-0.245,-0.242)	-0.021 (-0.250,-0.229)	-0.013 (-0.249,-0.236)	-0.015 (-0.246,-0.231)
Post in any OMHC	-0.009 (-0.244,-0.236)	-0.005 (-0.257,-0.252)	-0.021 (-0.262,-0.241)	-0.017 (-0.265,-0.248)	-0.013 (-0.262,-0.248)
Comment in the same OMHC	-0.007 (0.149,0.155)	0.004 (0.158,0.154)	-0.005 (0.149,0.154)	0.001 (0.146,0.146)	-0.001 (0.145,0.146)
Comment in any OMHC	0.001 (0.154,0.153)	0.003 (0.157,0.154)	-0.003 (0.152,0.154)	0.001 (0.150,0.149)	0.002 (0.147,0.145)

Table 5: The difference in the sentiment score of subsequent k comments/posts between “comment group”(former) and “no comment group”(latter), and their respective values.

Variable \ k	1	2	3	4	5
Post in the same OMHC	0.041** (-0.234,-0.275)	0.033* (-0.246,-0.279)	0.053** (-0.239,-0.292)	0.055** (-0.229,-0.284)	0.041* (-0.234,-0.275)
Post in any OMHC	0.038** (-0.255,-0.293)	0.031* (-0.261,-0.292)	0.047** (-0.257,-0.304)	0.051** (-0.251,-0.302)	0.041* (-0.263,-0.303)
Comment in the same OMHC	0.184*** (0.207,0.023)	0.153*** (0.202,0.049)	0.138*** (0.191,0.053)	0.125*** (0.187,0.063)	0.110*** (0.184,0.073)
Comment in any OMHC	0.169*** (0.205,0.036)	0.138*** (0.196,0.058)	0.127*** (0.189,0.062)	0.116*** (0.187,0.070)	0.104*** (0.183,0.079)

Table 6: The difference in the sentiment score of subsequent k comments/posts between “positive comment group” (former) and “negative comment group” (latter), and their respective values.

4.3 Implications for Design

This study informs that in online mental health communities, initially received social support may have significant impact on individuals’ future behaviors. That may result in chain reactions: received social support may affect an individual, and encourage him to provide others with social support; therefore more users will be affected, and the number of affected users may continue to increase when this process repeats. Therefore, it is beneficial for the whole community if each user is able to receive warm social support on the first post. To this end, platforms could identify and call attention to posts that have not received any responses, e.g., creating a special “unanswered posts” section and providing higher incentives for users who reply to these posts. Besides, setting up a response reminder mechanism and timely informing users of their received social support may further strengthen the chain effect of social support. In addition, considering the contagious effect of sentiment, the community could appropriately introduce robots to automatically add more positive comments under posts that receive too many negative comments.

5 Conclusion and Future Work

This study examined the effect of social support in the form of comments in online mental health communities. We focused on users in 22 subreddits and conducted causal analysis by matching users (1) who received social support with who did not receive social support; (2) who received positive social support with who received negative social support. The results show that social support is contagious. Those who received social support on the first post

would be more likely and quickly to post again and provide support for others. Also, those who received positive support would subsequently provide more positive social support. This study provides insights into how online mental health communities can serve as a better place for people to spread social support and improve mental health state.

One limitation we faced was the incomplete selection of mental health related subreddits. Although this study included main OMHCs in Reddit which were demonstrated in prior research, we noticed that there also existed some other OMHCs with fewer members. That might cause bias in our results such as the probability to return to post or comment in any OMHC. Also, we did not identified the difference among comments in features other than the sentiment. Future work could further conduct content analysis of comments to explore classification scheme of social support and study their effects respectively. Additionally, we only took into account the effect of social support on the first post. That is because it is difficult to control the similarity of users’ activities on multiple posts and separate the impact of social support of each post among multiple posts. In the future, we could explore a practical way to examine the marginal effect of social support on each post separately.

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